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Air Quality Forecasting using Foundation Models

A Spatio-Temporal Deep Learning Approach for Multi-Pollutant Forecasting in Delhi

Phalak Mehta S20240030393

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A Spatio-Temporal Foundation Model for Multi-Pollutant Forecasting

The Problem: Delhi experiences severe, recurring pollution episodes (PM2.5 > 250 µg/m³). Authorities need reliable 1 to 3-day forecasts to trigger early interventions (e.g., GRAP).

The Research Gap: Existing air quality models typically forecast a single pollutant at a single station. They fail to leverage the physical reality that pollutants travel with the wind (spatial relationships) and react with each other (chemical relationships). When sensors go offline, traditional models fail or rely on flawed pre-imputation.

WHAT are we building?

A deep learning foundation model (AQFM) that ingests irregularly sampled data from 37 stations and predicts multiple pollutants simultaneously up to 72 hours in advance.

It produces calibrated probabilistic forecasts, telling us not just what pollution will be, but how certain it is.

WHY this approach?

Foundation models offer massive scalability; a model trained on Delhi can be scaled to all over India without retraining from scratch.

Furthermore, unlike standard deep learning models, it can natively handle real-time streaming data that has missing values and missing variables, ensuring it doesn't break when sensors go offline.

HOW are we doing it?

1. Collect 8 years of CPCB station data & ERA5 weather.
2. Clean and unify into model-ready tensors.
3. Build an Axial Transformer with a Query-Based Decoder.
4. Train with realistic 'mixed masking' to simulate outages.
5. Evaluate against operational baselines like XGBoost.

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Our Data Sources

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We use two complementary datasets: ground-level measurements (CPCB) and satellite weather reanalysis (ERA5)

Source 1: CPCB Ground Stations

CPCB (Central Pollution Control Board) operates a network of automatic air quality monitoring stations across Delhi NCR. Each station measures pollutant concentrations and local weather every 15 minutes, 24/7.

Pollutants measured (9): PM2.5, PM10, NO, NO2, NOx, Ozone, SO2, NH3, CO
Weather measured (7): Temperature, Humidity, Wind Speed, Wind Direction, Solar Radiation, Pressure, Rainfall
Also measured: Benzene

PROBLEM: Sensors frequently go offline. Some stations have 15-35% missing data. Entire stations sometimes go dark for days or weeks. This is not random -- when one sensor fails, often ALL sensors at that station fail together (systemic block missingness).

37
Active Stations

2018-2025
Years Covered

~155M
Total Observations

Source 2: ERA5 Weather Reanalysis

ERA5 is produced by ECMWF (European weather agency). It combines satellite observations, weather station data, and physics-based models to produce the best estimate of 'what the weather actually was' at every point on Earth, every hour, going back to 1940.

We fetched ERA5 data for the Delhi region (2017-2025) and interpolated it to each of our 37 station locations. Key variables:
Temperature, Humidity, Wind (U+V), Pressure, Solar Radiation, Rainfall

Boundary Layer Height (BLH) -- THE critical variable
BLH tells us how high pollution can mix in the atmosphere. On winter mornings when BLH is low (~100m), pollution gets trapped at ground level. CPCB stations do NOT measure BLH. Only ERA5 provides it.

15 min
Temporal Resolution

0%
Missing Data

37 sites
Spatial Coverage

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37 Monitoring Stations Across Delhi NCR

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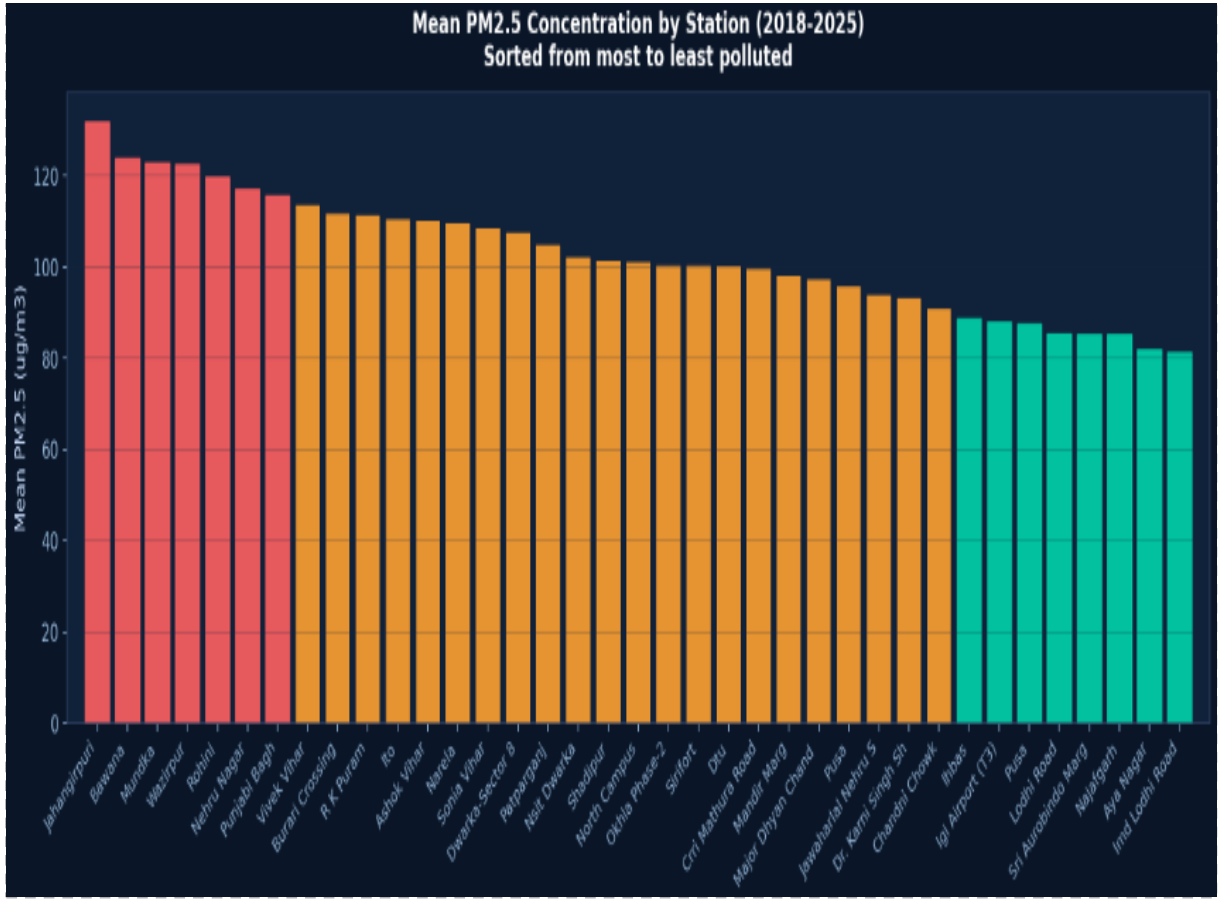
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Each dot is a station. Color shows how polluted it is on average (Mean PM2.5 concentration)



Dataset at a Glance

Stations	37 (operated by DPCC, IITM, CPCB)
Time Span	Jan 2018 -- Dec 2025 (8 years)
Sample Rate	Every 15 minutes (96 readings/day)
Timesteps	280,512 per station
Pollutants	PM2.5, PM10, NO, NO2, NOx, Ozone, SO2, NH3, CO
Met Variables	Temp, Humidity, Wind, Pressure, Rain, Solar Rad
Total Channels	17 measured (+2 derived = 19 for model)
Total Cells	176,442,048 (stations x channels x time)
Valid Obs	154,926,304 (87.8% observed)
Missing	21,515,744 (12.2% gaps)

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Exploratory Data Analysis

Understanding our data before we build anything

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EDA: PM2.5 Concentration Over 8 Years

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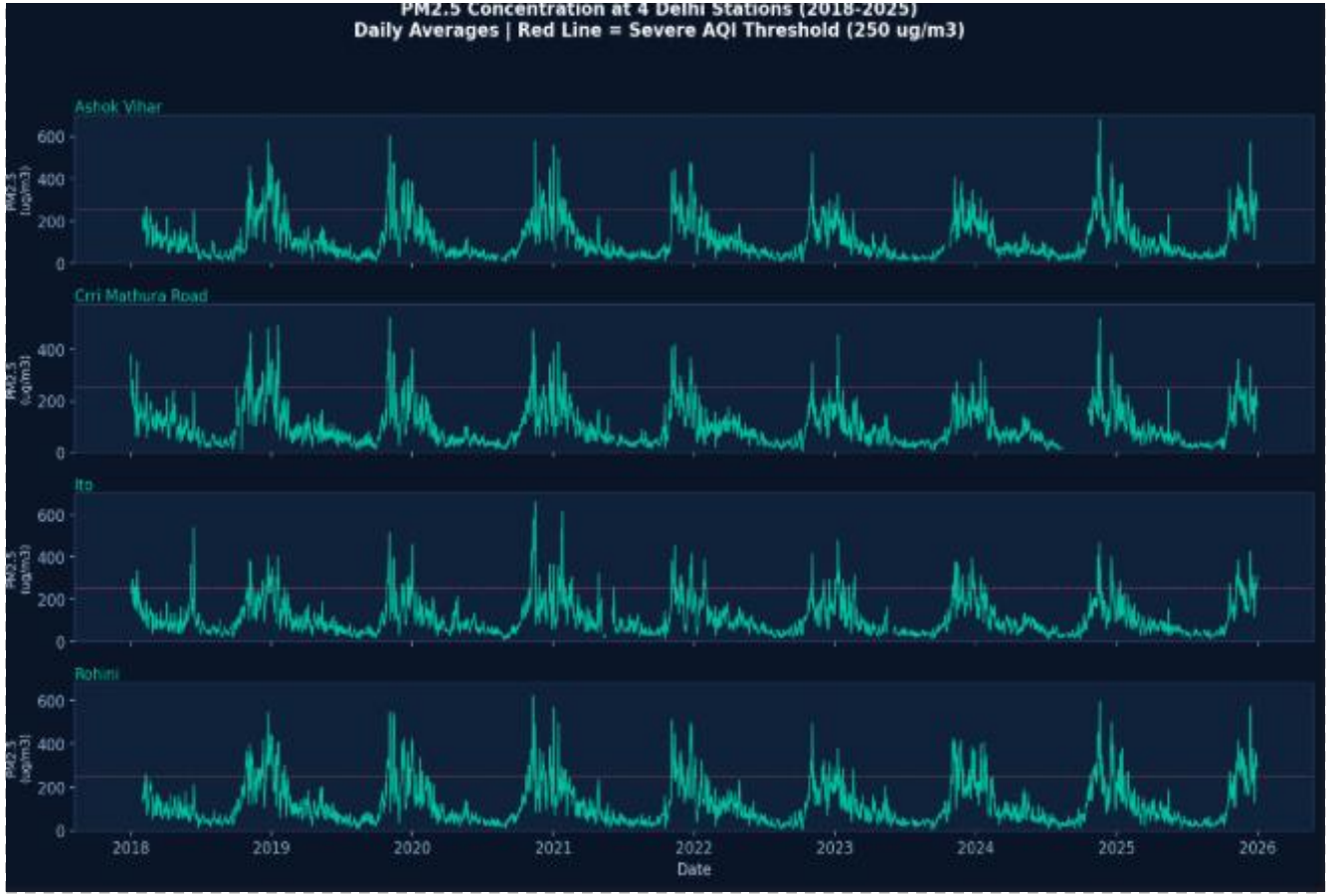
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Each graph shows daily PM2.5 levels at one station. Notice the dramatic spikes every winter (Oct-Jan).



What This Tells Us

Strong seasonal pattern: Every winter (Nov-Jan), PM2.5 rockets above 250 ug/m3.

Red line = 'Severe' AQI threshold (250 ug/m3). These are the dangerous episodes we need to predict.

Summers are relatively clean (30-80 ug/m3). But winters routinely hit 500-700+ ug/m3.

The pattern is consistent across all stations -- this means spatial modelling can help.

Key stats from data:

Mean PM2.5: 102.9 ug/m3
Median: 69.9 ug/m3
8.07% of readings > 250 (Severe)
0.67% of readings > 500 (Hazardous)

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EDA: The Daily Rhythm of Pollution

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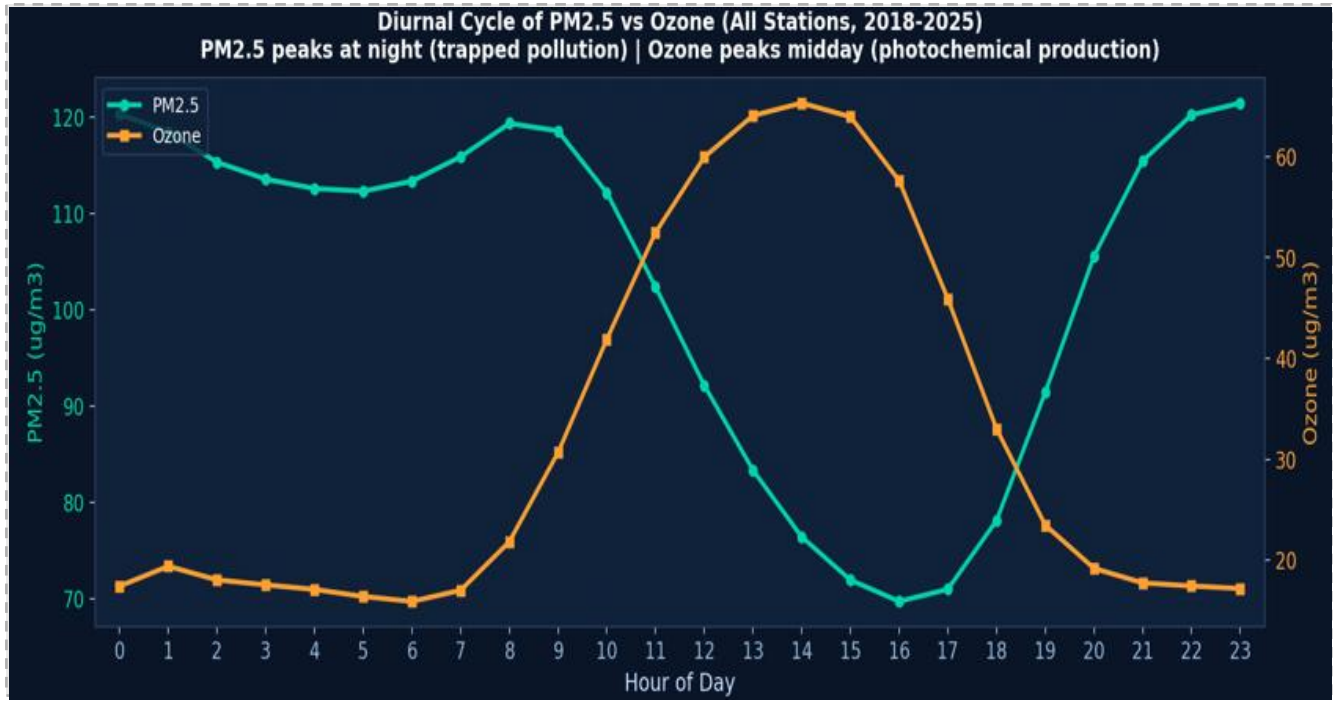
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Average pollutant concentrations by hour of day, computed from all 37 stations over 8 years of real data



What This Tells Us

PM2.5 (teal line) is highest at night and early morning, and lowest around 4-5 PM.

Ozone (orange line) shows the OPPOSITE pattern -- highest at midday. This is because Ozone is produced by sunlight reacting with NOx.

This is important because it means PM2.5 and Ozone are driven by different physics. A good model must understand both patterns.

The daily cycle is so strong that a simple 'same hour yesterday' baseline is surprisingly effective.

WHY THIS MATTERS FOR OUR MODEL: The model needs to learn the diurnal cycle (captured by Fourier time features in the tokeniser), AND the chemistry (captured by cross-channel attention in the encoder, so PM2.5 and Ozone can 'talk to each other' inside the model).

EDA: Why We Apply log(1+x) Transform

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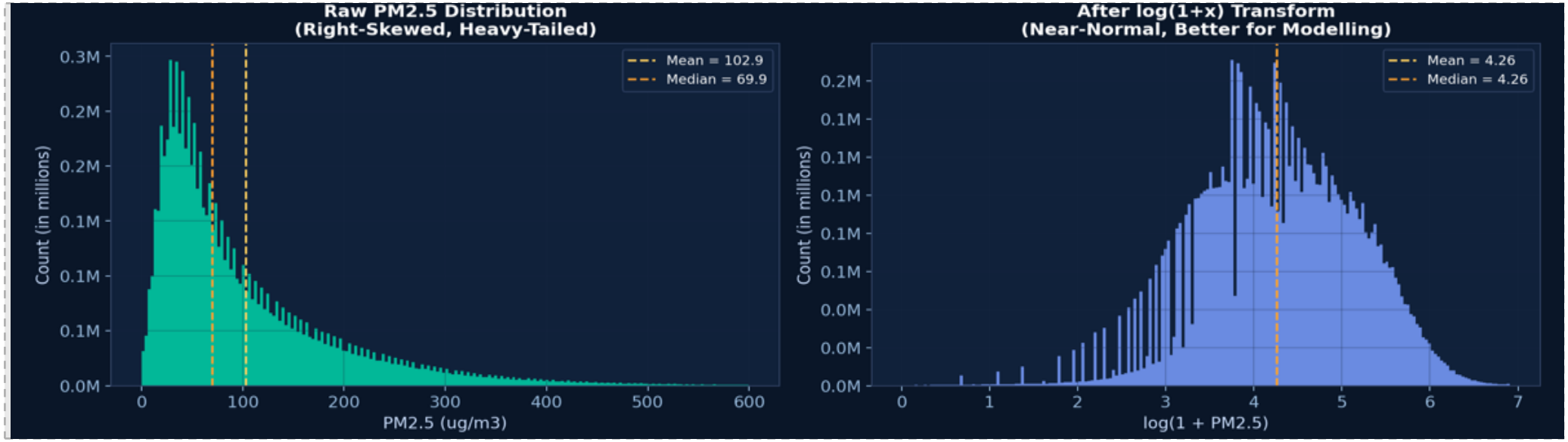
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Pollutant concentrations are heavily right-skewed. Applying log(1+x) makes them near-normal, which deep learning models learn much better.



The Problem (Left Plot)

Raw PM2.5 data has a long right tail -- most readings are low (median 69.9 ug/m3) but some reach 500-1000 ug/m3. The mean (102.9) is much higher than the median. Neural networks struggle with this kind of skewed data because the extreme values dominate the loss function and make training unstable.

The Solution (Right Plot)

After applying log(1+x), the distribution is nearly symmetric and bell-shaped. Mean and median are now very close (both ~4.26). This makes the data much easier for deep learning models to learn from. We apply this transform to ALL pollutant channels before training, and invert it after prediction to get real concentrations.

EDA: Cross-Pollutant Correlations

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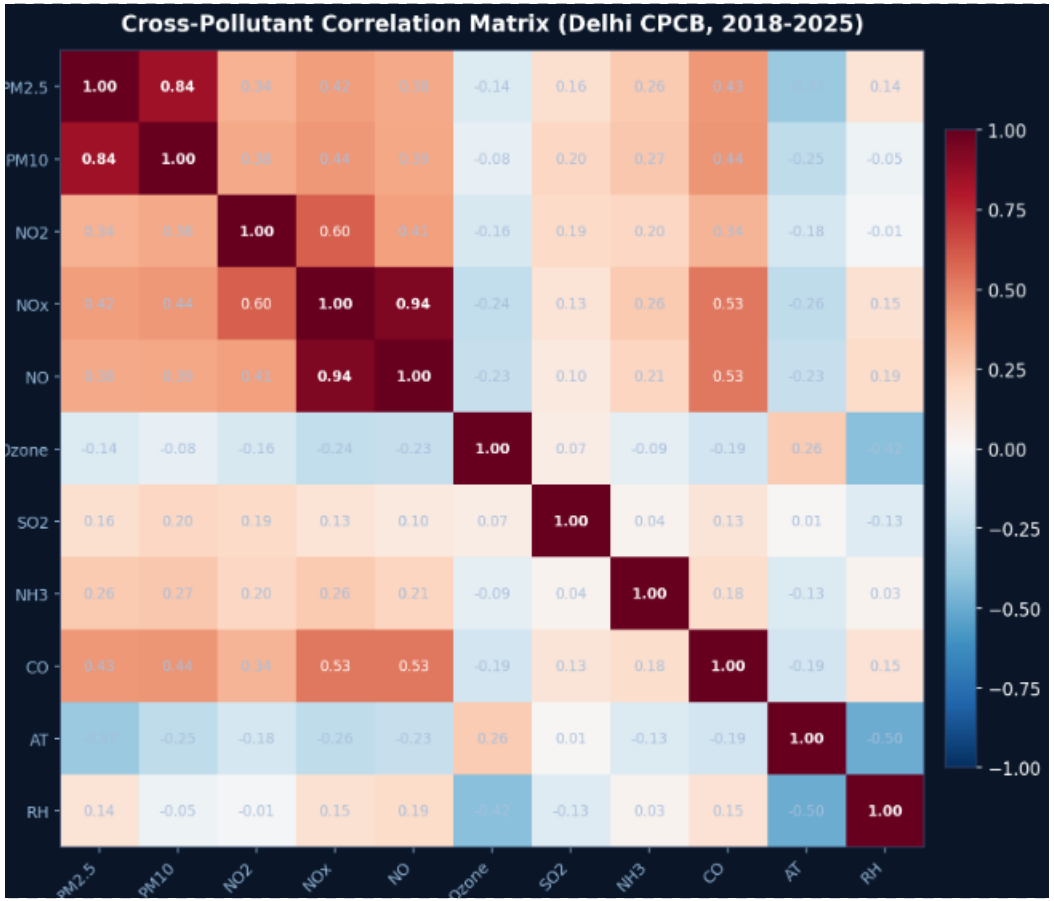
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This heatmap shows how each pollutant relates to every other. Computed from 1.84 million simultaneous observations across all stations.



Key Takeaways

1. PM2.5 correlates strongly with CO (r=0.65)
Both are products of combustion (vehicles, biomass).

2. NO and NH3 have a notable correlation (r=0.44)
Both are vehicular/agricultural emission markers.

3. Ozone positively correlates with CO (r=0.30)
But NEGATIVELY with PM10 (r=-0.17). Ozone is produced by sunlight, while particulates accumulate when there is no sunlight/wind.

4. AT (Temperature) is negatively correlated with PM10 (r=-0.38)
-- cold weather = more pollution.

WHY THIS MATTERS: These cross-pollutant relationships mean forecasting PM2.5 gets better if the model also sees CO, NOx, and weather. Our AQFM model learns all channels simultaneously. We also IMPOSE known physical laws (PM2.5 < PM10 always, NO + NO2 = NOx) directly into the model's architecture.

EDA: Data Quality - Missingness by Channel and Year

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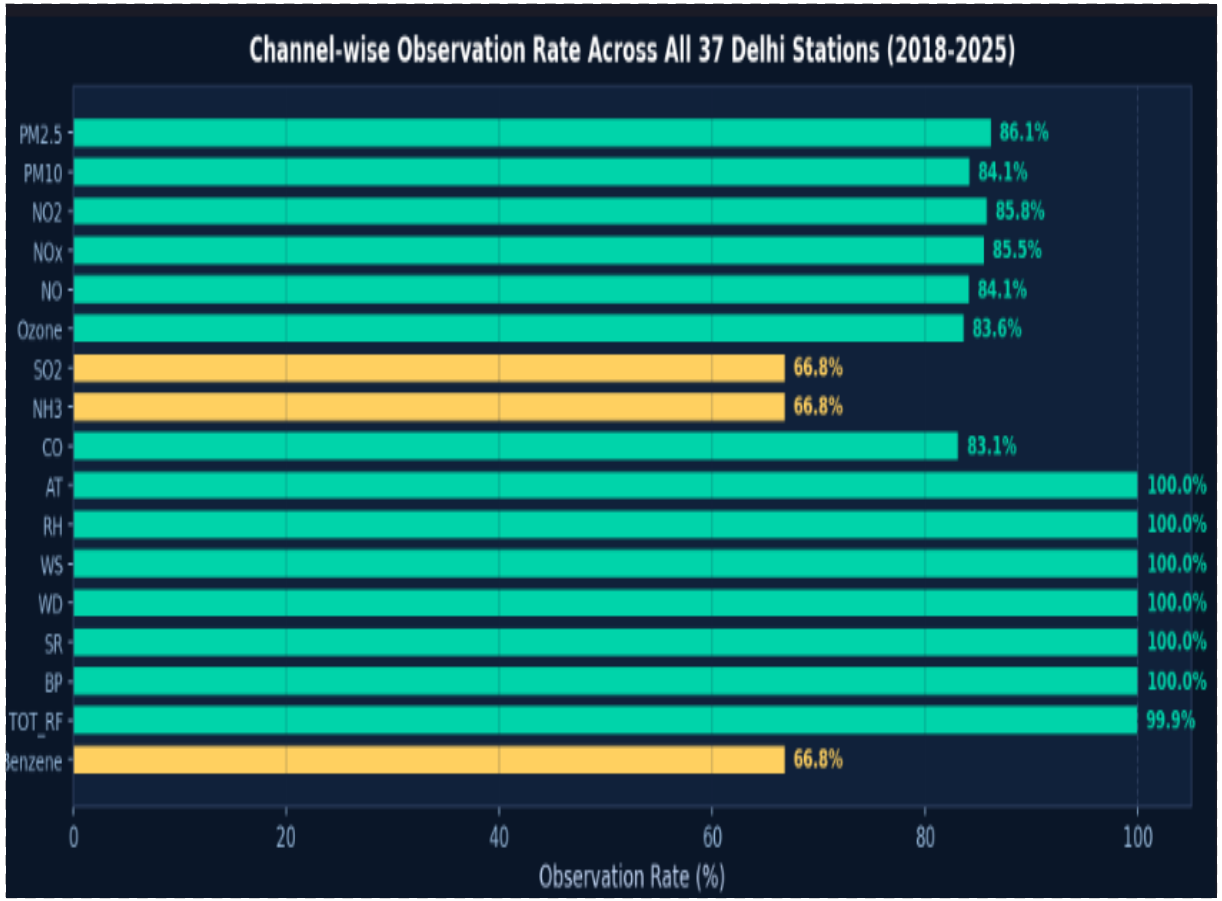
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This heatmap shows what percentage of readings are actually present for each channel, each year. Green = good coverage, yellow = partial.



What This Tells Us

Primary pollutants (PM2.5, PM10, NO2, etc.) have 75-93% coverage. That means 7-25% missing per year.

Coverage is IMPROVING over time -- 2024-2025 has 90%+ on most channels.

SO2, NH3, Benzene have worse coverage (~55-73%) because fewer stations measure them.

All weather channels (AT through TOT_RF) show 100% -- this is AFTER we filled their gaps using ERA5.

KEY DESIGN DECISION: We do NOT impute missing values. Because outages are 'systemic' (entire stations offline for days), imputation would just inject hallucinated data into our training signal. By handling missing data natively via binary masks, the model learns to rely on its neighbors (spatial attention) and weather (channel attention) to survive real-world sensor failures.

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Unifying CPCB and ERA5 Data for Model Ingestion

The 6-Step Preprocessing Pipeline

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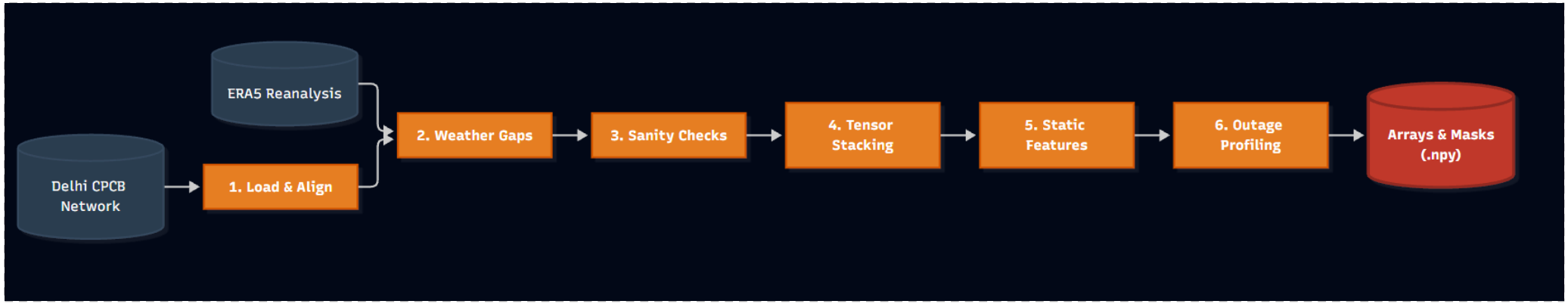
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We built a single script (preprocess_aqfm.py) that takes raw CSV files and produces model-ready numpy arrays



Step 1: Load & Align CSVs

300+ CSV files (one per station per year) are read, aligned to a single 15-minute timeline from Jan 2018 to Dec 2025, and stacked into one consistent grid. Duplicate timestamps are removed, negative readings are treated as missing.

Step 2: Fill Weather Gaps with ERA5

CPCB weather sensors (temperature, humidity, wind, etc.) have significant gaps. ERA5 reanalysis fills these gaps perfectly -- all 7 meteorological channels are now at 100% coverage. 22+ million individual gap values were filled.

Step 3: Physical Sanity Checks

Clip physically impossible values: e.g., negative concentrations, temperature outside [-15, 55] C, wind speed > 50 m/s. Also flag PM2.5 > PM10 violations (0.44% of data) which the model handles via its physical constraint architecture.

Preprocessing: Steps 4-6 and Final Output

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The remaining steps produce the exact arrays the model expects as input

Step 4: Stack Into Dense Arrays

All data from all 37 stations and 17 channels is stacked into two large tensors: 'values' and 'observed' (a binary mask). Shape: (37 stations, 17 channels, 280512 timesteps).

Step 5: Compute Static Features

For each station, we compute 4 fixed properties: (1) average PM2.5, (2) average PM10, (3) elevation proxy from pressure, (4) station density (stations within 5km).

Step 6: Fit Outage Distribution

We fit a LogNormal distribution to 125,506 real outages (median 0.5 hours), so training simulates realistic missing-data patterns, not artificial ones.

Final Output Files (saved in preprocessed/ folder -- these are what the model will consume)

File	Shape	Size	Description
values.npy	(37 x 17 x 280,512)	706 MB	All pollutant and weather readings. Missing values = 0.
observed.npy	(37 x 17 x 280,512)	706 MB	Binary mask: 1 = real measurement, 0 = missing.
timestamps_h.npy	(280,512)	2.2 MB	Hours since Jan 1, 2018. Model uses this for time encoding.
coords.npy	(37 x 2)	< 1 KB	Latitude and longitude of each station.
static.npy	(37 x 4)	< 1 KB	Static features per station (mean PM, density, etc.)

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Baseline: Zero-Shot Forecasting with IBM Granite TTM

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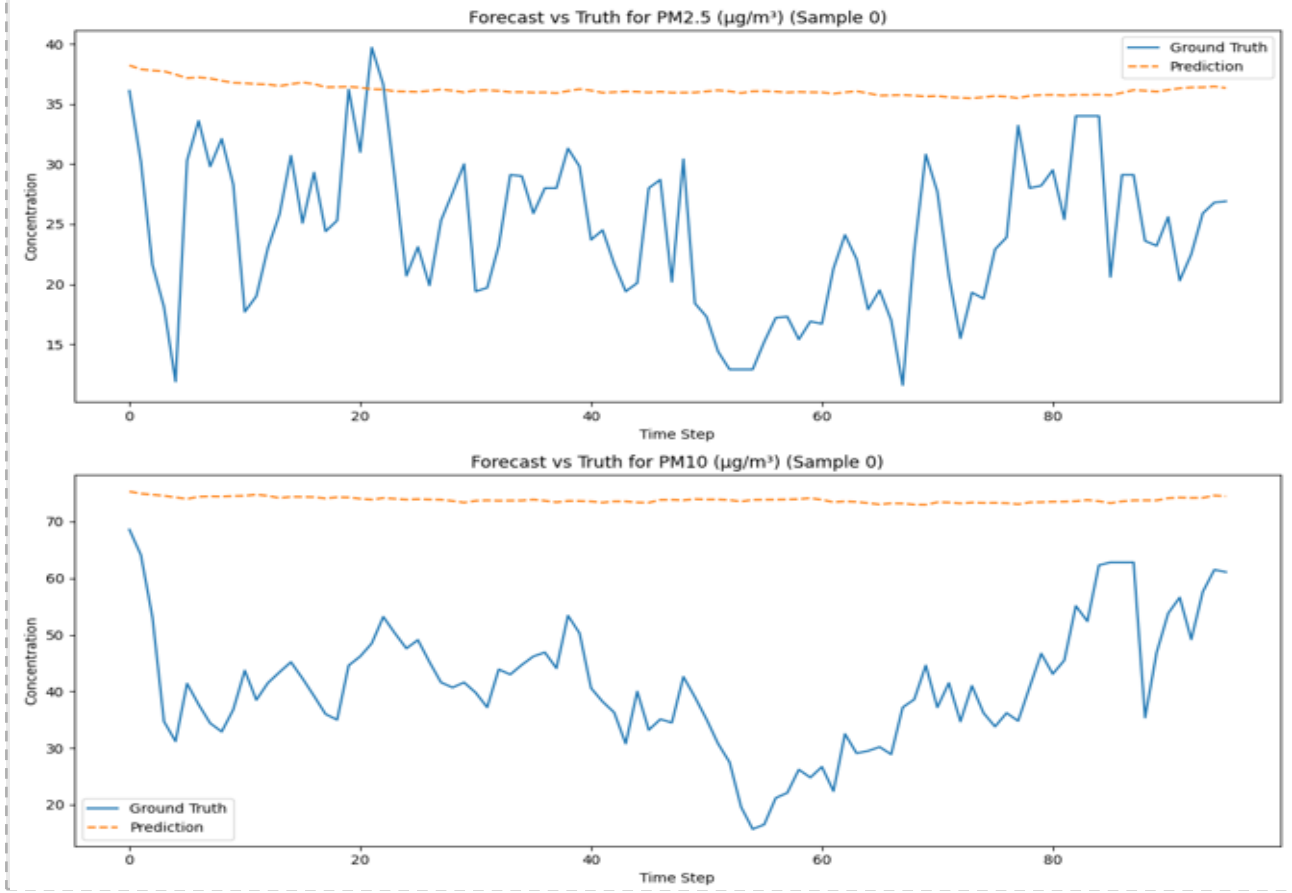
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Before building our own model, we tested an existing pre-trained time-series foundation model on our data to establish a baseline performance.



What We Did

We used IBM's Granite Time-Series Tiny Time Mixer (TTM), a pre-trained foundation model for time-series forecasting. We ran it in ZERO-SHOT mode -- meaning we did NOT train or fine-tune it on our data at all. We just gave it 21 days of history and asked it to forecast the next 4 days.

32.6 ug/m3
PM2.5 Mean Absolute Error

61.8 ug/m3
PM10 Mean Absolute Error

Key Observations

The model correctly learns the diurnal (daily) cycle even without being trained on this specific dataset.

BUT it fails completely on the magnitude scale. It wildly overpredicts all concentrations, forecasting values way above extreme events for almost all data points.

This proves off-the-shelf time-series models fail zero-shot on air quality. We need AQFM: a custom model designed to understand realistic concentration limits and physical constraints.

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The AQFM Model

What we are building next

(Air Quality Foundation Model — Spatio-Temporal Transformer)

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AQFM Architecture -How the Model Works

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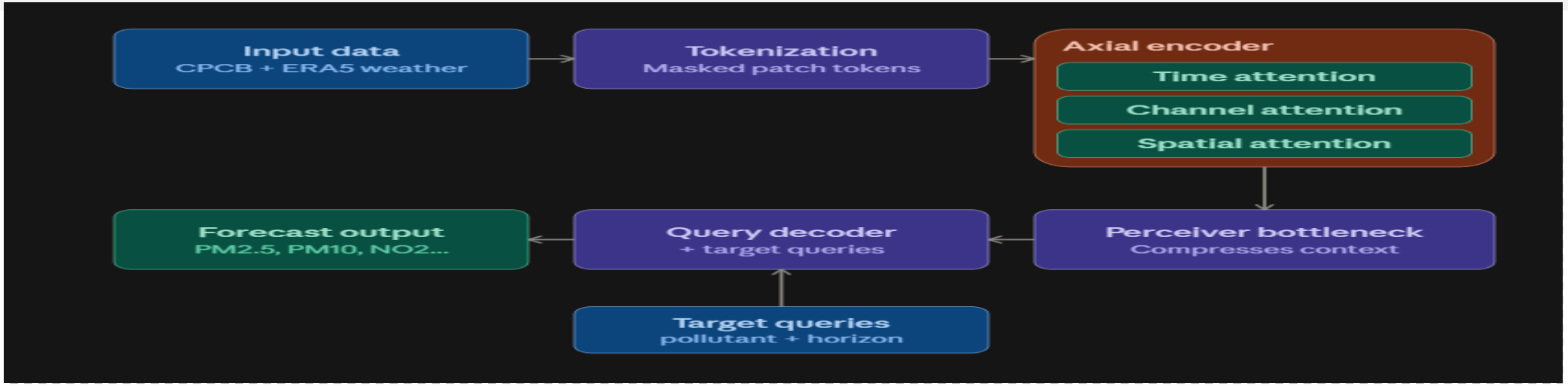
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1. Tokenisation

Each 4-hour window of a single pollutant at a single station becomes one 'token'. Missing data is handled by including a mask alongside each token -- the model knows exactly what is real vs missing.

2. Axial Encoder

The encoder runs attention in three directions: across TIME, across CHANNELS, and across SPACE. This is much more efficient than attending to everything at once.

3. Query-Based Decoder

We give the decoder a 'query': which pollutant, at which station, how far into the future? This means we can forecast any subset of pollutants at any horizon.

4. Probabilistic Output

The model outputs a probability distribution (mixture of 3 log-normal distributions) -- not just a value, but a confidence range. Critical for operational decisions.

Key Design Decisions — Why AQFM Is Different

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Three things that make this model specifically suited for air quality, based on what we learned from the EDA

Physical Constraints Built Into the Architecture

We discovered that PM2.5 is always a subset of PM10 (by physical definition), and NO + NO2 always equals NOx (by chemistry).

Instead of hoping the model learns these rules from data, we ENCODE them directly: the model predicts PM10 and then a fraction (0 to 1) to get PM2.5. This makes PM2.5 > PM10 physically IMPOSSIBLE.

Similarly, the model predicts NOx and a split-fraction for NO vs NO2. Zero additional parameters. Zero nonsense output.

Wind-Aware Spatial Attention

A station 8km UPWIND is far more informative than one 8km DOWNWIND. Standard models only use distance between stations. Ours adds the bearing relative to the wind direction as an attention bias.

This means: when the wind blows from the northwest, the model automatically pays more attention to northwest stations. This is a physics-informed inductive bias that distance-only graphs cannot express. The AQFM design spec calls this the 'wind-aware spatial bias MLP' and it operates per attention head.

Training with Realistic Missing Data Patterns

From our EDA, we know that data gaps are SYSTEMIC -- when one sensor fails, all sensors at that station fail together for hours or days.

During training, we simulate three types of missingness: (1) mask the future (50% -- this IS forecasting), (2) mask entire stations for realistic durations fitted from real outage data (35%), and (3) drop random channels (15% -- forces the model to forecast PM2.5 from wind and neighbouring stations alone).

Training Plan & Model Configuration

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Two-phase training curriculum: first learn general patterns with mixed masking, then fine-tune for specific forecast horizons

Model Configuration

Context Window	5.3 days (512 timesteps at 15-min)
Forecast Horizon	Up to 72 hours (288 timesteps)
Patch Size	4 hours (16 timesteps per token)
Neighbourhood	12 nearest stations per sample
Model Dimension	384
Attention Heads	8
Axial Encoder Blocks	4 (= 12 attention layers)
Decoder Layers	4 (self-attn + cross-attn + FFN)
Mixture Components	3 (log-normal mixture)
Optimiser	AdamW, lr=3e-4, weight decay 0.05
Normalisation	Masked RevIN (context-only stats)
Total Parameters	~23 million

Phase 1: Mixed-Masking Pre-Training

The model trains on a mixture of three masking strategies:
Tail masking (50%): hide the last K patches -- this is pure forecasting.
Block masking (35%): simulate station-wide outages (duration from fitted LogNormal).
Channel masking (15%): drop random channels -- forces multi-modal learning.

This teaches the model both forecasting AND imputation simultaneously. The masking weights are designed so the model sees forecasting tasks most often, since that is our primary goal.

Phase 2: Horizon Fine-Tuning

After the model has learned general patterns, we fine-tune specifically for the forecast horizons we care about operationally:
1 hour -- real-time monitoring and alerts
6 hours -- same-day planning
24 hours -- next-day school closure decisions
72 hours -- GRAP stage activation planning

Only tail masking (pure causal forecasting). Lower learning rate (5e-5). Strict causal split: no future information leaks into the context window.

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How We Will Evaluate the Model

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Standard metrics like MAE are misleading for air quality. We use metrics that matter for real-world operational decisions.

A pooled yearly MAE looks good because clean summer months (PM2.5 = 30-50) dominate the average and hide the model's failures during the dangerous winter episodes (PM2.5 = 300-600) that the forecast ACTUALLY EXISTS to predict. We therefore stratify ALL metrics by AQI band and season.

Metrics We Will Report

- Skill stratified by AQI band (Good / Moderate / Poor / Severe / Hazardous)
- Episode detection: can the model predict PM2.5 > 250 ug/m3 at 24h and 48h lead?
- Calibration: are the model's 90% and 95% prediction intervals actually correct?
- CRPS by lead time: where does skill decay to climatology (random guessing)?
- Imputation scored separately from forecasting (different tasks, different failures)

Baselines We Must Beat

Persistence: Predict tomorrow = today. Floor; anything below this is broken.

Seasonal Naive: Predict = same hour yesterday. Captures diurnal cycle for free.

Climatology: Average value for this station, hour, and month. Long-horizon competitor.

Gradient-Boosted Trees: XGBoost on lagged features. Known to be VERY strong on this task.

IBM TTM (our baseline): Zero-shot result we already have. PM2.5 MAE = 32.6 at 96h.

PatchTST / iTransformer: State-of-art time-series models. Isolates value of spatial attention.

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Summary & Current Status

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Work Completed

- ✓ Fetched all CPCB data (37 stations, 2009-2025)
- ✓ Fetched ERA5 reanalysis (2017-2025, interpolated to stations)
- ✓ Comprehensive EDA: seasonal patterns, correlations, missingness
- ✓ Zero-shot baseline with IBM TTM (PM2.5 MAE=32.6 at 96h)
- ✓ Unified CPCB+ERA5 (all met channels now at 100% coverage)
- ✓ Full preprocessing pipeline: 6-step script producing model-ready .npy files
- ✓ All files and work uploaded to shared Drive

What's Next

- ▸ Build the AQFM tokeniser (patch tokens + time encoding)
- ▸ Build the axial encoder (time + channel + space attention)
- ▸ Build the query-based decoder (Design B)
- ▸ Implement loss functions (NLL + CRPS with order-statistic trick)
- ▸ Implement physical constraint module (PM2.5<PM10, NO+NO2=NOx)
- ▸ Two-phase training on preprocessed data
- ▸ Full evaluation against all baselines

Thank You